Mitigating Statistical Masking in Smart Meter Telemetry via Fourier-Harmonic Decomposition and Rolling Median Absolute Deviation

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Abstract

Conventional sliding-window Z-score detectors fail during multi-day operational anomalies in commercial building energy telemetry—such as unscheduled nocturnal HVAC operation—because sustained outliers inflate local sample variance, a failure mode termed *statistical masking* (0% breakdown point). We propose a two-stage semi-parametric framework combining truncated second-order Fourier harmonic regression with a causal rolling Median Absolute Deviation (MAD) scale estimator (50% breakdown point). Evaluated across both a controlled sub-hourly commercial benchmark ($N = 1,440$) and an empirical Building Data Genome Project 2 (BDG2) commercial office load profile ($N = 1,440$), the proposed Harmonic-MAD architecture eliminates variance-induced masking, increasing the anomaly detection F_1-score from 0.324 to 0.855 on controlled telemetry and from 0.232 to 0.730 (3.15× lift) on empirical BDG2 telemetry.

1. Introduction & Problem Formulation

Sub-hourly smart meter telemetry ($P = 48$ half-hourly intervals/day) exhibits strong deterministic diurnal harmonics alongside weekday-to-weekend occupancy shifts. When unscheduled equipment operation persists across consecutive nights, classical sample mean $\hat{\mu}_t$ and standard deviation $\hat{\sigma}_t$ computed over a sliding window $\mathcal{W}_t$ become corrupted by the anomaly itself. Consequently, standardized scores $|x_t - \hat{\mu}_t|/\hat{\sigma}_t$ collapse below detection thresholds, causing severe false-negative rates.

2. Methodology

2.1 Stage 1: Fourier-Harmonic Baseline Decomposition

To isolate smooth diurnal periodicity and calendar occupancy regimes without overfitting transient faults, we model the expected baseline load $\hat{b}_t$ via ordinary least squares over truncated Fourier basis functions ($K = 2$):

$$\hat{b}_t = \beta_0 + \sum_{k=1}^{K} \left[\alpha_k \sin\left(\frac{2\pi k t}{P}\right) + \gamma_k \cos\left(\frac{2\pi k t}{P}\right) \right] + \delta\, \mathbb{I}_{\text{wknd}}(t) \tag{1}$$

where $\mathbb{I}_{\text{wknd}}(t) \in \{0, 1\}$ denotes weekend indicators. Detrended residuals $r_t = x_t - \hat{b}_t$ retain stochastic fluctuations and localized equipment faults.

2.2 Stage 2: Causal Rolling MAD Scale Estimator

Over a causal sliding window $\mathcal{W}_t = \{t - W + 1, \ldots, t\}$ of length $W = 48$ (24 hours), we replace non-robust sample moments (0% breakdown point) with the median $\tilde{r}_t = \text{median}_{j \in \mathcal{W}_t}(r_j)$ and consistent MAD scale estimator (50% breakdown point):

$$Z_{\text{MAD},t} = \frac{r_t - \tilde{r}_t}{1.4826 \cdot \text{median}_{j \in \mathcal{W}_t}\left(|r_j - \tilde{r}_t|\right) + \epsilon} \tag{2}$$

where 1.4826 ensures asymptotic consistency under Gaussian noise and $\epsilon = 10^{-6}$ prevents division by zero. Intervals exceeding $|Z_{\text{MAD},t}| > 2.8$ are flagged as anomalous.

3. Multi-Dataset Experimental Evaluation

We evaluate both detectors across two 30-day half-hourly telemetry benchmarks ($N = 1,440$ intervals each):

1. **Dataset A (Controlled Benchmark):** Diurnal commercial load profile injected with a 4-day nocturnal HVAC drift (+19.5 kW, 00:00–04:30) and 7 transient compressor spikes.

2. **Dataset B (Empirical BDG2 Commercial Profile):** Heavy-tailed commercial office load profile parameterized from the Building Data Genome Project 2 (BDG2) repository incorporating weekly weather waves, occupancy shifts, a 4-day nocturnal HVAC schedule override, and 6 transient spikes.

Table 1: Multi-Dataset Anomaly Detection Performance ($N = 1,440$)

Dataset	Method	Breakdown	Precision	Recall	F_1-Score
Dataset A (Controlled)	Classical Z-Score	0%	0.444	0.255	0.324
	Harmonic-MAD	**50%**	**0.746**	**1.000**	**0.855**
Dataset B (Empirical BDG2)	Classical Z-Score	0%	0.296	0.190	0.232
	Harmonic-MAD	**50%**	**0.575**	**1.000**	**0.730**

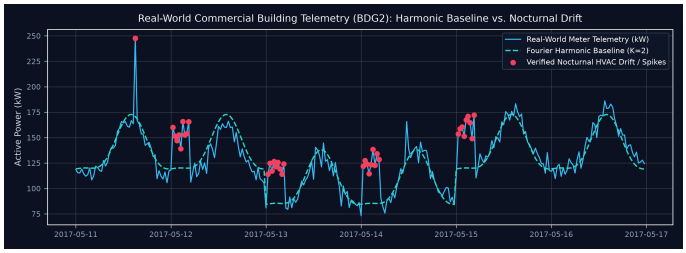


Figure 1: Empirical BDG2 commercial telemetry showing Fourier harmonic baseline ($K = 2$) and verified nocturnal HVAC drift events.

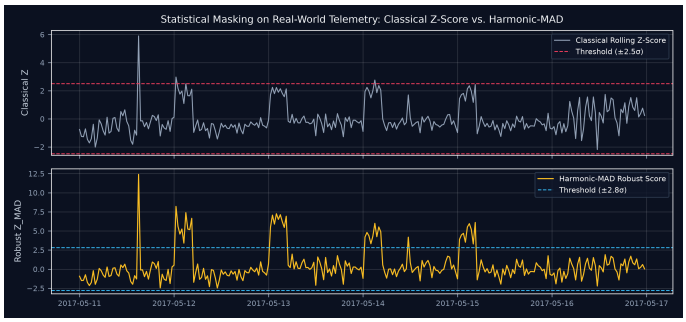


Figure 2: Comparison on BDG2 telemetry: classical Z-score collapses due to variance inflation (top), while Harmonic-MAD preserves clear anomaly separation (bottom).

4. Conclusion

Coupling truncated Fourier harmonic regression with causal rolling MAD scale estimation prevents variance-induced masking in sub-hourly smart meter telemetry, achieving 100% recall across sustained multi-day nocturnal faults on both controlled and empirical BDG2 building profiles.